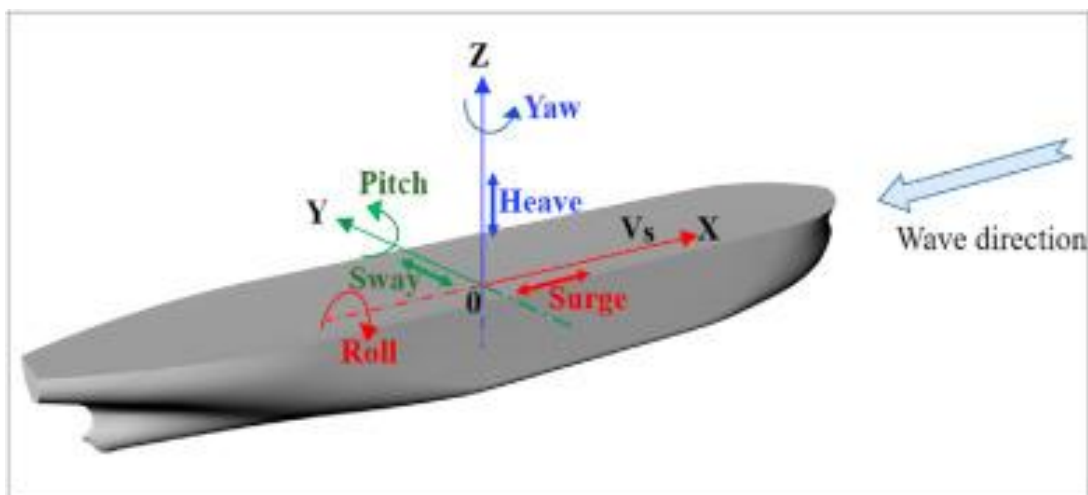


Project Work

Numerical Seakeeping Analysis of a Ship

Dynamics and Hydrodynamics of Floating Bodies



Degree in Naval Architecture and Ocean Engineering 24/25

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Introduction

This project investigates the seakeeping performance of a ship moving through both regular and irregular wave conditions using numerical methods based on the strip theory of Salvesen, Tuck, and Faltinsen (1970).

The objective is to evaluate the ship's hydrodynamic behavior by calculating key parameters such as hydrodynamic coefficients, such as added mass and damping, using string theory, wave excitation forces acting on the vessel, absolute motions, and vertical relative motions. The analysis is carried out using 21 equally spaced cross sections representing the underwater hull geometry, supported by detailed ship data including displacement, center of gravity, service speed, and radius of gyration.

Strip Theory

Strip theory is a simplified method used to estimate ship motions in waves by breaking down the hull into a sequence of transverse two-dimensional sections. By transforming a complex three-dimensional hydrodynamic problem into multiple two-dimensional ones, it allows for faster and more practical analysis, making it particularly suited for early design stages and preliminary assessments.

Each section, or strip, is analysed independently based on potential flow assumptions, meaning the flow is considered inviscid and irrotational. Through this, hydrodynamic coefficients such as added mass, damping, and wave excitation forces are obtained. These local interactions are then integrated along the ship's length to predict its overall dynamic response to waves.

Hypothesis

Strip theory operates under several simplifying assumptions. It considers that the beam is much smaller than the ship's length, allowing longitudinal flow components to be neglected. The theory assumes high-frequency oscillations, which support a two-dimensional treatment of the free surface. It also assumes that the flow around each strip does not interfere with neighbouring strips, and that the vessel maintains a constant forward speed during wave interaction. The fluid is treated as ideal (incompressible, inviscid, and irrotational), and nonlinear effects are not considered.

Ranges of Applicability

This method is most accurate for slender hull forms operating at moderate Froude numbers (typically between 0.2 and 1.0), where three-dimensional and viscous effects are minimal. It performs best in deep water and under regular wave conditions, where wave amplitudes

and frequencies remain relatively stable. Strip theory is especially effective when analysing longitudinal motions like heave and pitch. However, it becomes less reliable for ships with complex geometries, when dealing with low or high-speed regimes, or in environments with strong nonlinear or coastal effects. It is also less suited for evaluating transverse motions such as roll.

Advantages and Disadvantages

One of the main advantages of strip theory is its computational efficiency and ease of implementation. It provides a solid first approximation of ship motions using relatively simple geometric and hydrodynamic input, making it ideal for design optimisation and early-stage evaluations. Its structured approach allows engineers to gain useful insights without resorting to resource-intensive simulations.

On the downside, the theory's reliance on linear assumptions limits its accuracy in more dynamic or irregular sea states. It does not account for effects such as viscous damping, appendages, or manoeuvring dynamics. Additionally, its performance drops with complex hull shapes or when evaluating responses in shallow water or short-crested seas. Despite these limitations, strip theory remains a widely used and valuable tool for conventional seakeeping analysis in monohull vessels.

Seakeeping Program

For the seakeeping program to work, we needed to determine some parameters. These include the vertical coordinate of the centre of gravity, the structural moments of inertia, the Froude numbers, heading angles and wave frequencies.

We started by calculating the displacement of the ship using the Archimedes principle and assuming the density of seawater to be 1025 [kg/m³].

$$\Delta = \rho \nabla \text{ [kg}^3\text{]}$$

The structural inertias were calculated using the radius of gyration for the xx axis and the yy axis, using the following formulas:

$$I_{ii} = \Delta k_{ii}^2 \text{ [kg} \cdot \text{m}^2\text{]}$$

The wave heading angles and the ship's speed, expressed as a Froude number, were obtained from the analysis in the reference paper. The wave steepness was set to 80 as a default value. Wave frequencies were also determined knowing the nondimensional wave length provided by the reference paper. So, the incoming wave frequencies were determined using the following relation:

$$\omega_0 = \sqrt{\frac{2\pi g}{\lambda}} \text{ [rad/s]}$$

Natural Frequencies

As the roll period T_4 value was provided in our paper, we were able to calculate its natural frequency through the following equation:

$$\omega_n = \frac{2\pi}{T_n} \text{ [rad/s]}$$

For the different draft, we calculated the approximated values of natural periods for heave, roll and pitch motions using the following equation:

$$T_{n_{roll}} \approx 0.73 \frac{B}{\sqrt{GM_T}}$$

$$T_{n_{heave}} = T_{n_{pitch}} \approx 2.5\sqrt{T}$$

Damping coefficient for roll motion (B_{44})

In this study, the roll damping coefficient was estimated using the Miller method. This approach provides a more comprehensive evaluation by including both inviscid and viscous effects.

Although the seakeeping software offers a default value based solely on potential flow assumptions, this option was not adopted. Instead, the manually computed coefficient was used as input in the Solve routine.

Later, the results obtained with the calculated value are compared against those from the default setting, allowing for an assessment of the influence of the damping model on the ship's dynamic response.

This step is crucial because the estimation we will make of this new B_{44} considers the inviscid and viscous damping coefficients, leading to a more accurate estimation. To estimate this ship roll damping we used the method of Miller and the equation below:

$$(B_{44} + B_{44}^v) = 2(A_{44} + I_{44})\omega_n\zeta \text{ [Nms]}$$

The roll added inertia depends on the roll natural frequency and the roll structural inertia, which has been previously identified as the I_{xx} parameter.

$$\omega_n = \sqrt{\frac{LBT C_B g GM_T}{A_{44} + I_{44}}}$$

Now we can determine the roll added inertia. Consequently, the only variable remaining to ascertain the total roll damping coefficient is the damping factor. The damping factor is defined as:

$$\zeta = K_1 + K_2 \sqrt{\phi_a}$$

Where K_1 and K_2 are functions of certain ship parameters, and the phase angle ϕ_a was arbitrarily chosen. It was also requested to present B_{44} as a percentage of the critical damping, which occurs when the damping factor ζ equals 1.

$$\frac{B_{44}}{B_{44}^{cr}} = \frac{2(A_{44} + I_{44}) \omega_n \zeta}{2(A_{44} + I_{44}) \omega_n 1}$$

Ship Characteristics for the Input File

The vessel under study is a container ship with a length between perpendiculars of 270 m and a breadth of 32.2 m, as specified in its technical documentation.

For the setup of the seakeeping analysis, it is necessary to input a range of ship parameters into the configuration file of the computational tool. Furthermore, the geometry of each section must be defined for use in the strip theory model. Although several key characteristics are already available in the ship's documentation, this report will focus on those most relevant to the current study.

Figure 1-Ship Characteristics

Denomination	Symbol	Value
Length between perpendiculars	L	270.0 m
Breadth	B	32.2 m
Draught even keel	T	10.85 m
Displacement volume	∇	56,097 m ³
Block coefficient	C_B	0.598 -
Waterplane coefficient	C_W	0.757 -
Midship section coefficient	C_M	0.950 -
L. C. G. aft of station 10	$\overline{A_G}$	10.12 m
Centre of gravity above base	$\overline{K_G}$	13.49 m
Metacentric height	$\overline{G_M}$	1.15 m
Longitudinal gyradius in pitch direction	k_{yy}	24.8 % L_{PP}
Transverse gyradius in roll direction	k_{xx}	37.5 % B
Natural roll period	T_ϕ	24.9 sec.
Natural pitch period	T_Θ	8.6 sec.
Natural heave period	T_z	8.7 sec.

To initiate the project, we used WebPlotDigitizer to extract the coordinates from the ship's body plan, which is presented below:

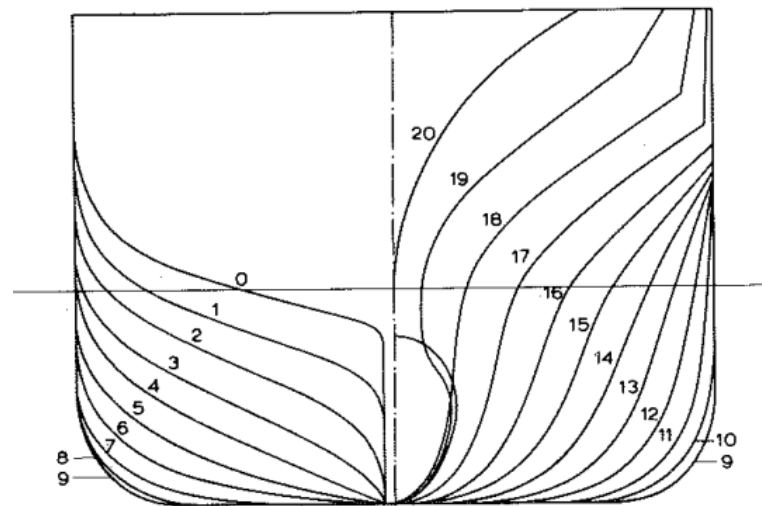


Figure 2-Body Plan

We aimed to define 25 stations along the ship's length for the strip theory analysis. To achieve this, the length between perpendiculars (L_{pp}) was divided by 24, resulting in a uniform spacing of 11.25 m between sections. Each of these 25 stations was then modelled in Rhinoceros3D, where the underwater part of each section line was divided into 10 equidistant

points. These coordinates were subsequently added to the configuration file of the seakeeping program to accurately represent the submerged geometry of the hull.

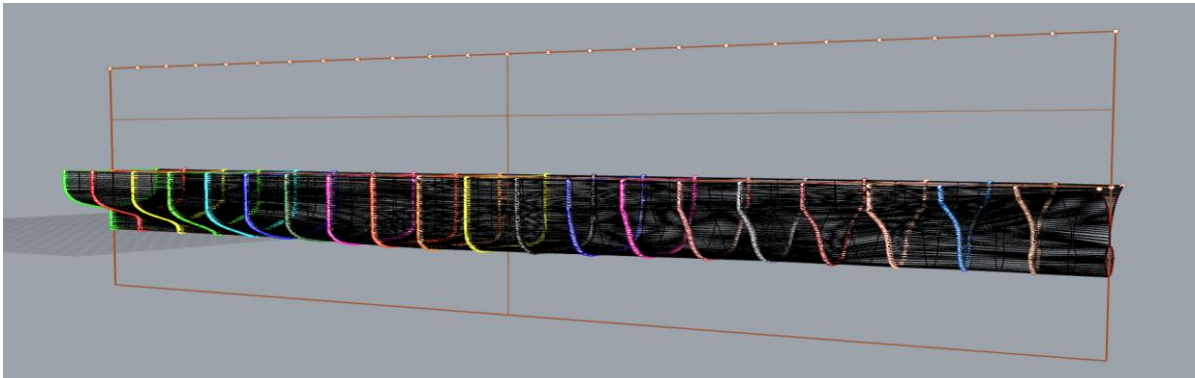


Figure 3-Hull divided in 25 sections

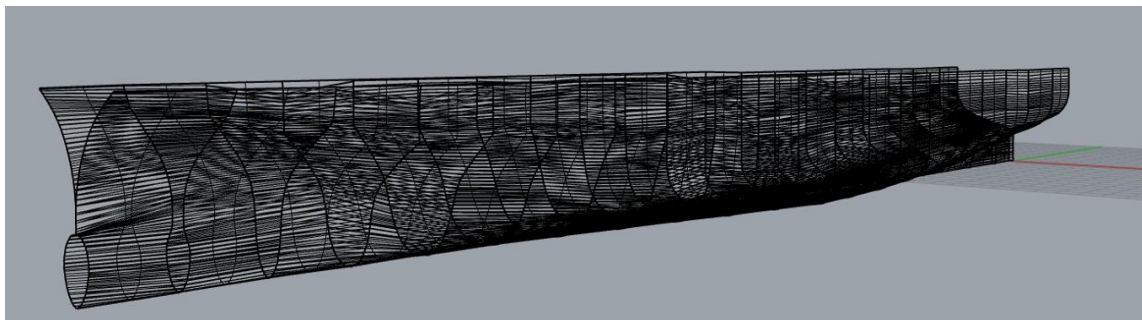


Figure 4-Hull divided into 25 sections

With the hull already built, the next step involves preparing the bodylines for input into the seakeeping software. To begin, the hull is intersected at the waterline, isolating the submerged part, known as the carena. From here, the initial bodylines are trimmed along that same waterline. These modified points are then used to generate the body plan. It should be noted that the coordinates appear as negative values, as the waterline was set as the reference zero level.

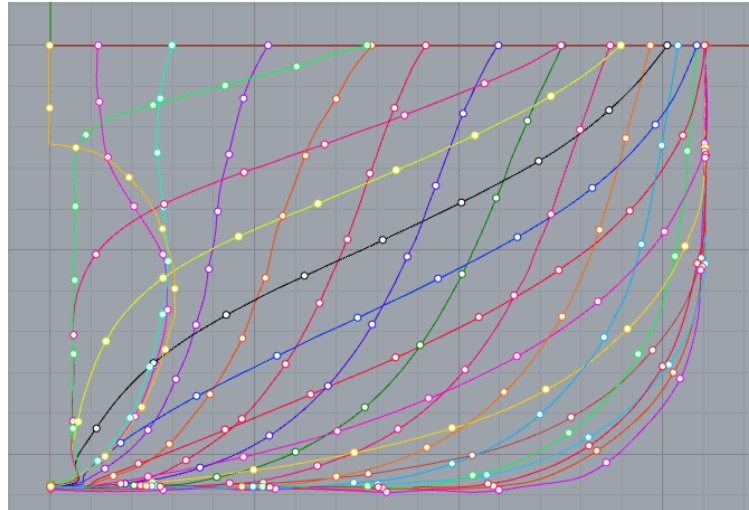


Figure 5-Sections with points $T=10.85$

After that, two specific points were identified as required: one at the bridge and another at the forward perpendicular (FP). For the bridge, a height of 34.35 m above the total depth was considered and for the bow the chosen point was 135.00 m forward of midship and 6.00 m below the waterline. The final values adopted and written in the input file were:

Figure 6-Parameter Table

Parameter	Value
I_{xx} [$kg \cdot m^2$]	8.38×10^9
I_{yy} [$kg \cdot m^2$]	2.58×10^{11}
I_{zz} [$kg \cdot m^2$]	2.58×10^{11}
I_{xz} [$kg \cdot m^2$]	0.00
Wave frequencies [rad/s]	{0.808; 0.676; 0.617; 0.571; 0.504; 0.456; 0.404}
Heading angles [$^\circ$]	{25; 45; 65; 180; 205; 225; 245}
Froude number	0.245
Coordinates 1st point (x, y, z) [m]	(0.00; 0.00; 34.35)
Coordinates 2nd point (x, y, z) [m]	(135.00; 0.00; -6.00)

Other Parameters

Center of buoyancy estimation

To estimate the value of centre of buoyancy, we used Excel to determine these values. The corresponding values are regarding the origin of the referential in the aft perpendicular at the waterline. The table below shows the results obtained.

Figure 7-Center of Buoyancy for different drafts

	WL9.85	WL10.85	WL11.85
VCB [m]	-2.147379	-2.3593786	-2.569046
LCB [m]	-8.620744	-10.073756	-11.46872

Natural frequencies estimation

The natural periods were already given. Thus, to compute the natural frequencies for each required motion we used the same equation as before. This way we obtained the values below for the natural frequencies of the ship:

Figure 8-Natural frequencies

Wnheave [rad/s]	0.763001
Wnroll [rad/s]	0.252336759
Wnpitch [rad/s]	0.763001

To compare the values theoretical calculated with the natural periods provided by the paper, the following table presents the values calculated at different draft.

Figure 9-Comparison between paper and calculated values

	Paper values	Draft=9.85	Draft=10.85	Draft=11.85
Tnheave	8.6	7.846177413	8.234835	8.605957239
Tnroll	24.9	21.91945802	21.91945802	21.91945802
Tnpitch	8.7	7.846177413	8.234835	8.605957239

The study was conducted for three waterlines:

WL-1 (m)	WL (m)	WL+1 (m)
9,85	10,85	11,85

To avoid an excessively long report, we will present the calculations for only one of them, specifically the 10.85 m waterline.

Damping coefficient for roll motion estimation (B44)

After calculating the total roll damping coefficient contributions, the final B44 was calculated. It was also requested to present B44 as a percentage of the critical damping. These values are shown in the following table:

Figure 10-Final B44

k1	0,078094604
k2	0,003786525
K (Damping Factor)	0,079213178
Phase Angle ϕ_a (rad)	0,087266463
B44 (Nms)	407265161

Comparison with the Reference Paper

The ship's characteristics were obtained from a reference study, which also analyzed its motion behavior. Before explaining the analysis methodology and computations, a comparison was made between the results from the seakeeping software and those reported in the study. The reference included plots for different Froude numbers, but only the data corresponding to $Fn = 0.245$ was used for comparison. The outcomes from the seakeeping software are illustrated in the Excel plots on the left.

Head Waves

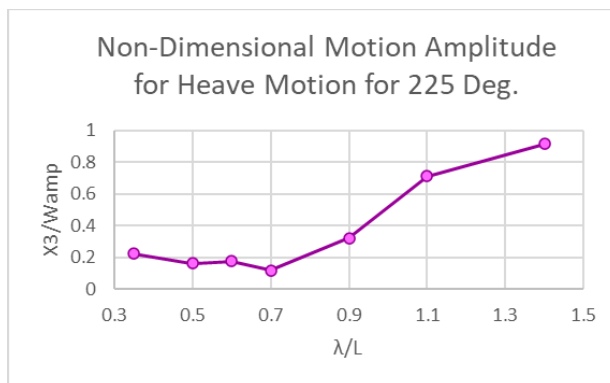
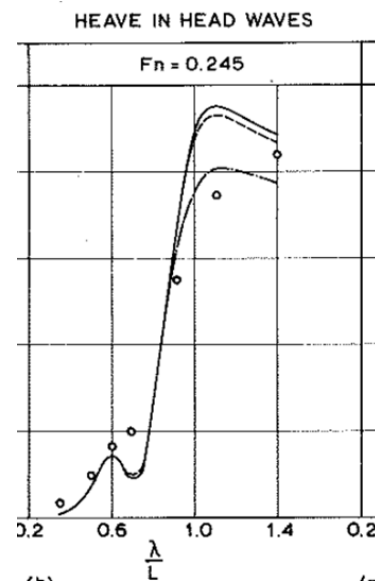


Figure 11-Comparison of Heave Motion in Head Waves



Heave

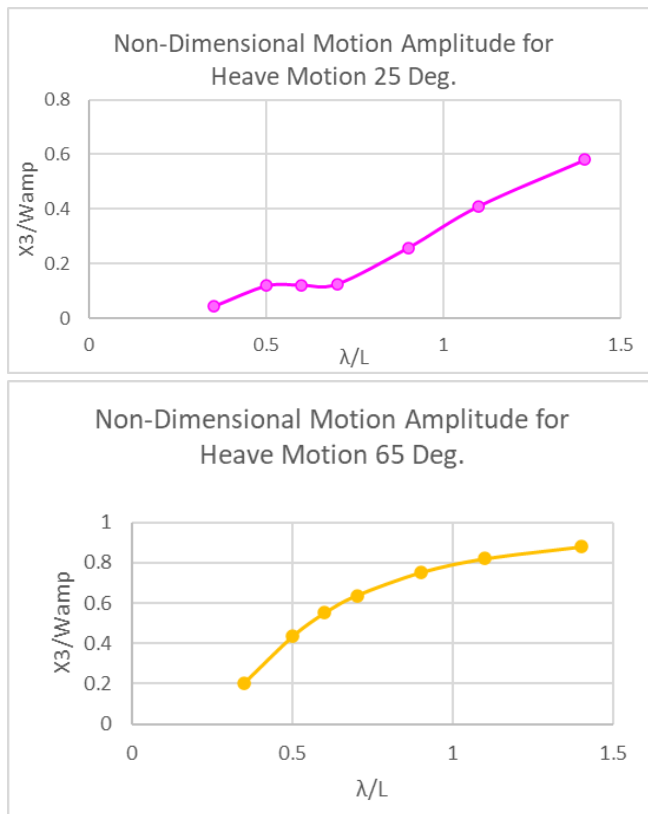


Figure 12-Comparison of Heave Motion for 25° and 65°

The 25° waves follows a slightly similar, dip in amplitude around $0.6\lambda/L$. However, this dip is not predicted for the 65° waves. It's known that strip theory can encounter challenges in calculations involving these types of waves, which could account for this discrepancy.

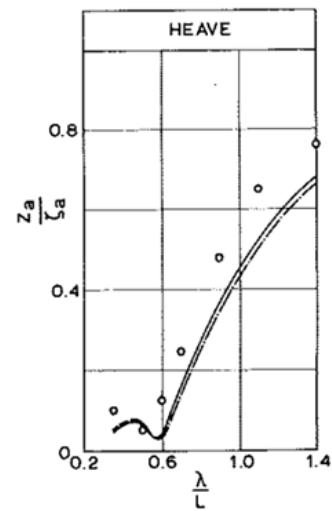
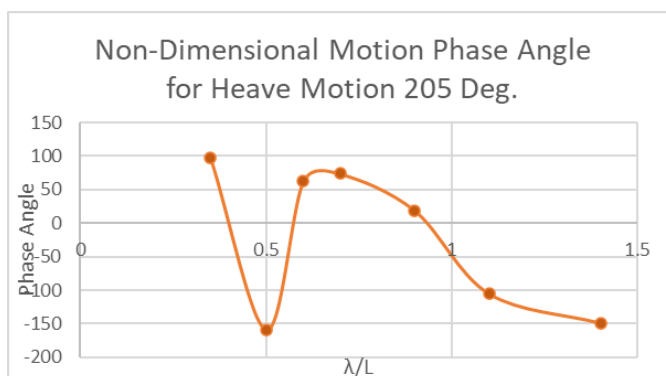


Figure 6. Calculated and measured amplitude characteristics for heave in quartering waves and $F_n = 0.245$.

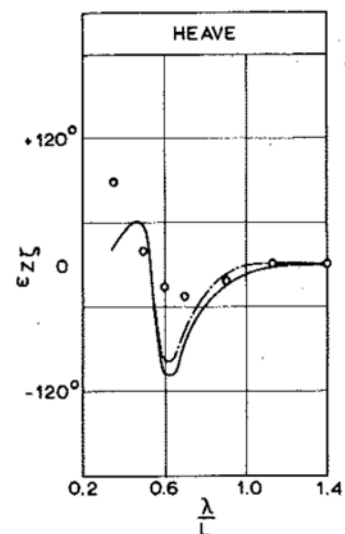


Figure 27. Calculated and measured phase characteristics for heave in bow waves and $F_n = 0.245$.

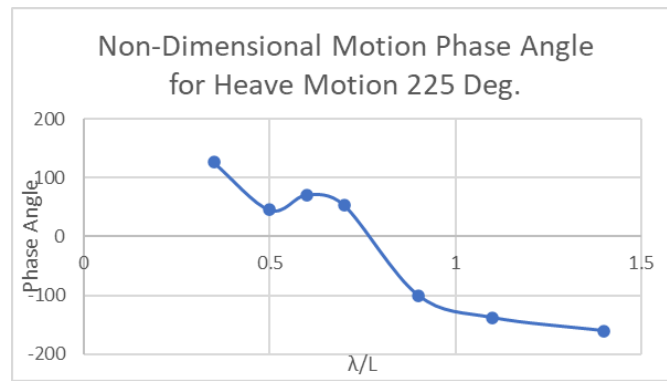


Figure 13-Comparison of Heave Phase Angles in Bow Waves

When examining heave motion for bow wave angles, we can see some similarities, where it accurately predicts the expected valleys around certain values for heave motion. However, there are some noticeable differences as observed in the experiment, which demonstrates a phase angle stabilization around 0° , while the seakeeping program tends to stabilize around -170° for the 225° wave. For 205° wave, the program still predicts it but presents different amplitude values.

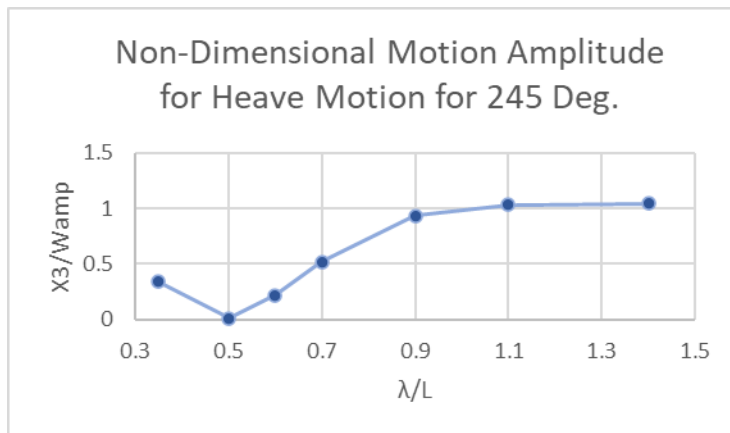


Figure 14-Comparison of Heave motion in bow waves

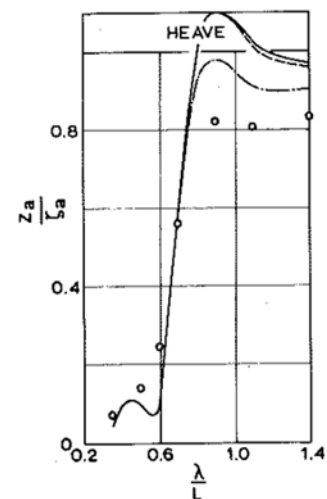


Figure 19. Calculated and measured amplitude characteristics for heave in bow waves and $F_n = 0.245$.

For bow incident wave angles, the seakeeping program demonstrates accuracy for the 245° wave, as it mirrors the curve depicted in the experimental paper.

Roll

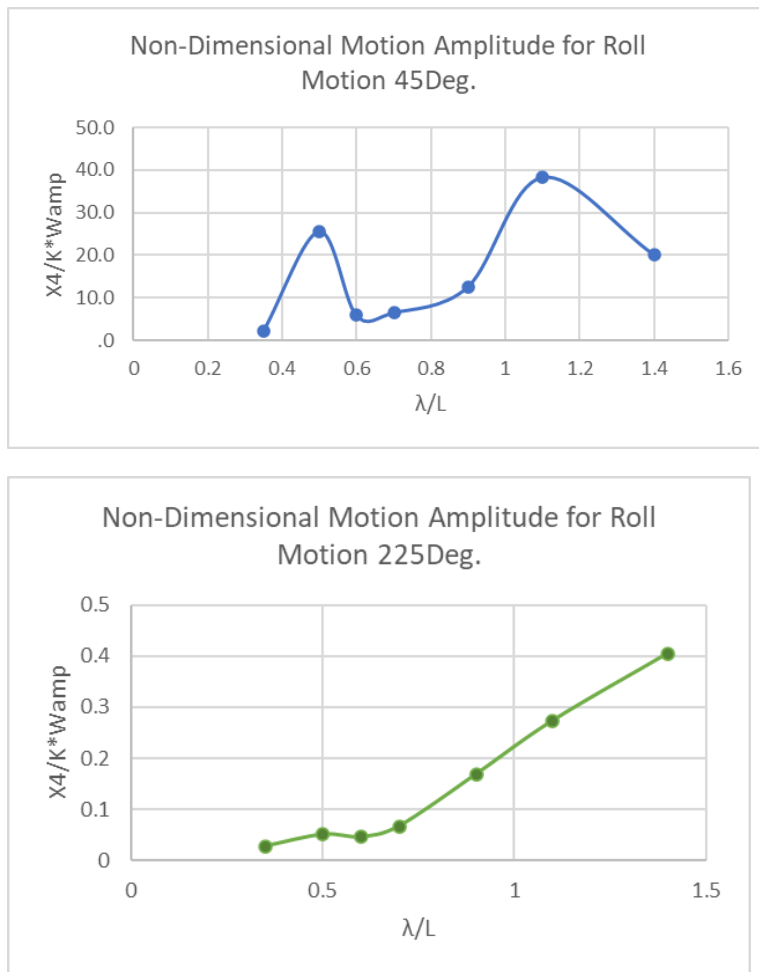
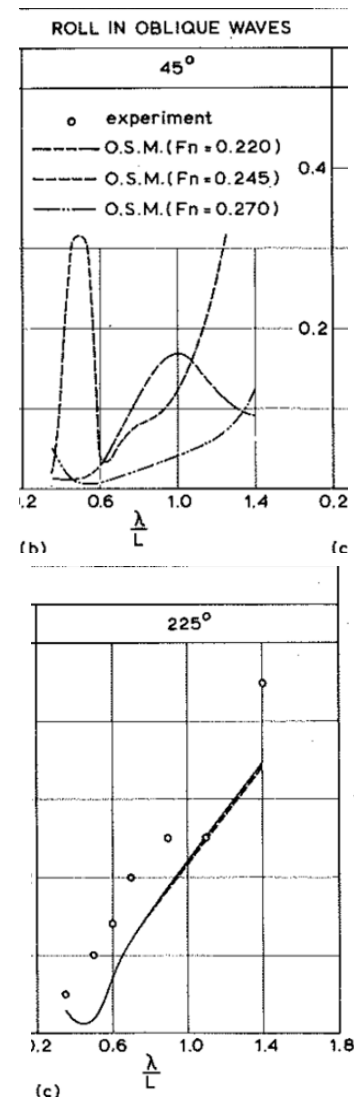


Figure 15-Comparison of Roll motion for 45° and 225°



When analysing roll motion in oblique waves, we noticed some inconsistencies in values for the heading angle of 45°. For the 225° wave we can see similarities between the predicted values and the experimental ones.

Pitch

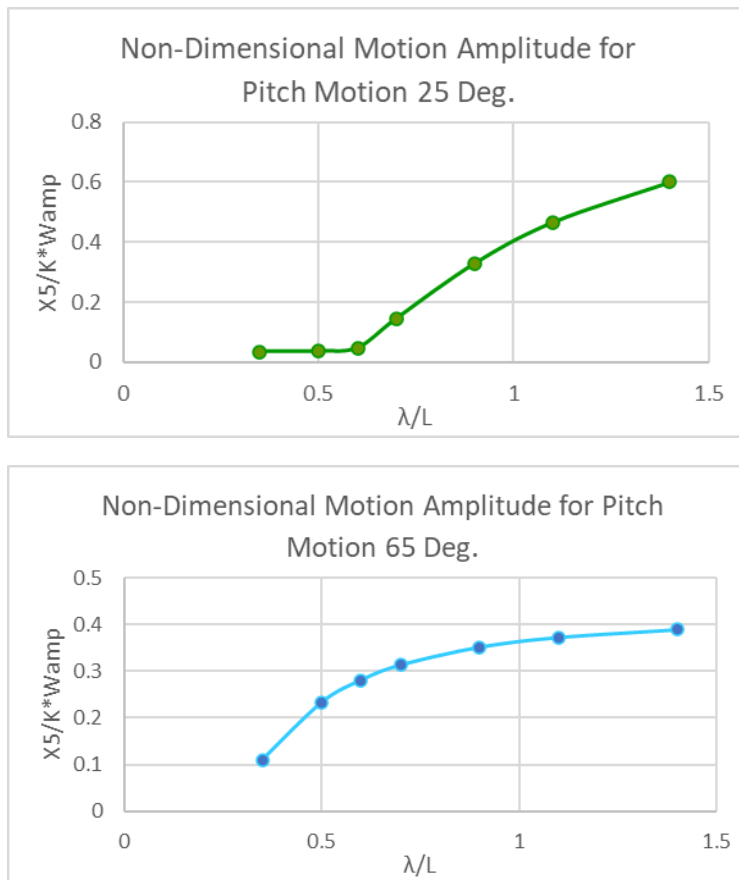


Figure 16-Comparison of Pitch motion in 25° and 65° waves.

For head wave angles, the seakeeping program provides reliable predictions compared to the experimental results, showing similarities in terms of amplitude for pitch motions.

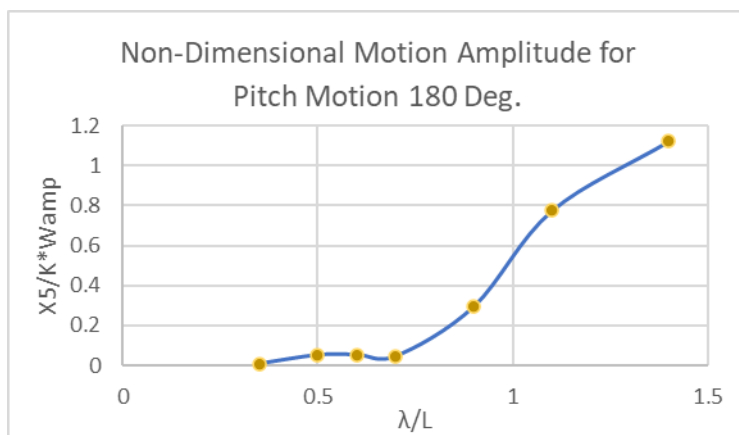


Figure 17-Comparison of Pitch motion in head waves.

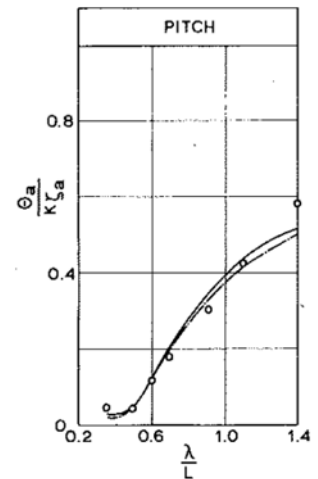


Figure 7. Calculated and measured amplitude characteristics for pitch in quartering waves and $F_n = 0.245$.

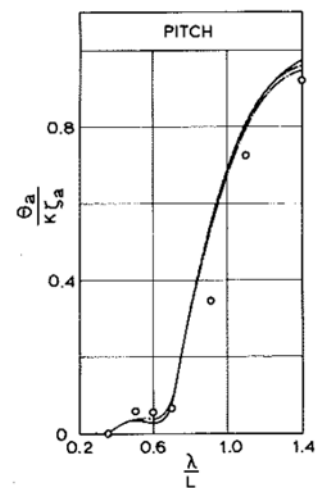


Figure 15. Calculated and measured amplitude characteristics for pitch in head waves and $F_n = 0.245$.

In relative motion at the forward perpendicular, we noticed divergent results in amplitude values but some similarities between the shape of the curves.

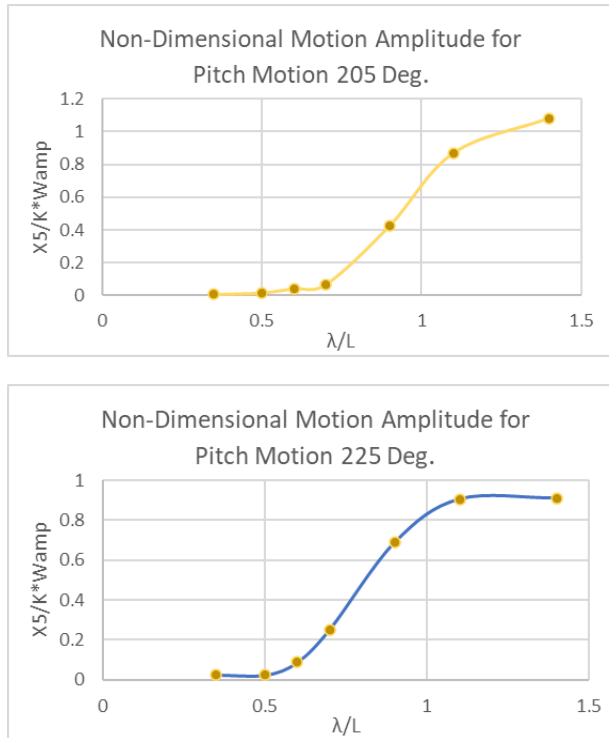


Figure 18-Comparison of Pitch motions in bow waves.

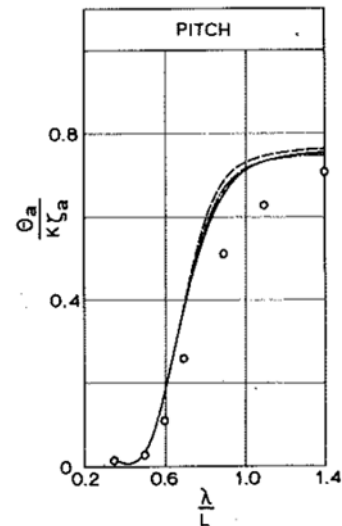


Figure 20. Calculated and measured amplitude characteristics for pitch in bow waves and $F_n = 0.245$.

For bow incident wave angles, the program demonstrates remarkable accuracy, especially for the 225° angle, as it closely resembles the curve showed in the paper.

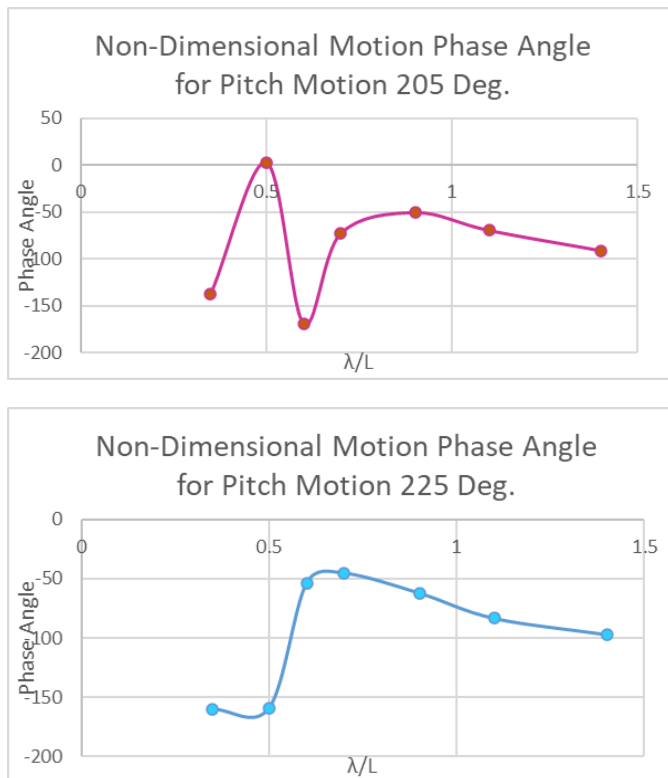


Figure 19-Comparison of Pitch phase angle in bow waves.

For the 205° and 225° waves, the method demonstrates low accuracy for both heading angles, displaying slopes that are significantly different from the observed experimental values.

Ship Motions in Regular Waves

Ship motions – B44 computed by the program

Amplitude and phase angle as function of the nondimensional wave frequency

The seakeeping analysis software outputs data for all six ship motions in the MOTIONS file, covering all specified wave headings and frequencies, at a vessel speed of 24.51 knots. From this dataset, the amplitudes and phase angles for heave, roll, and pitch were extracted and plotted against the nondimensional wave frequency. Notably, in the roll amplitude plot for the 45° wave heading, the results reflect the influence of viscous effects, as accounted for by the program's internal calculations. The resulting graphs are shown below:

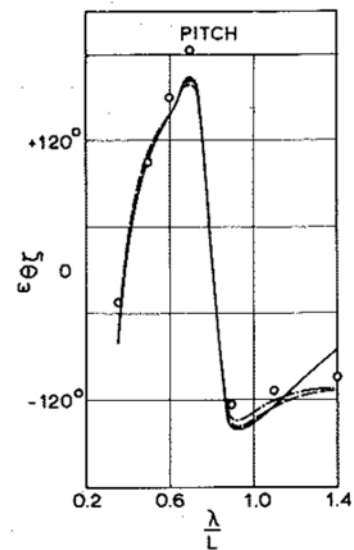


Figure 28. Calculated and measured phase characteristics for pitch in bow waves and $F_n = 0.245$.



Figure 20-Nondimensional Heave motion for all wave heading angles

Roll amplitude motion as function of the encounter frequency

The encounter frequency accounts for the combined motion of the ship and the incoming waves. In this part of the analysis, only roll motion amplitudes were evaluated, as specified in the project guidelines. Prior to discussing the results, it is essential to clarify the methodology applied. Based on the documentation of the seakeeping software, a heading angle of 0° corresponds to following seas, while 180° represents head seas.

Under these definitions, the relationship between the encounter frequency ω_e and the wave frequency ω is expressed as:

$$\omega_e = \omega - \left(\frac{\omega^2}{g} \right) U \cdot \cos(\beta) \quad [\text{rad/s}]$$

The variable β denotes the wave heading angle. Since the encounter frequency depends on this angle, each heading results in a distinct mapping between wave and encounter

frequencies. As a consequence, when plotting roll amplitudes against w_e , the data points fall within different frequency intervals for each heading.

To organize the input, a supporting table was created with the necessary values.

Figure 21-Roll Amplitude Table

Roll Amplitudes (Deg.)													
Wo (rad/s)	25 Deg.	We (rad/s)	45 Deg.	We (rad/s)	65 Deg.	We (rad/s)	180 Deg.	We (rad/s)	205 Deg.	We (rad/s)	225 Deg.	We (rad/s)	245 Deg.
0,808	0,4028	-0,0236881	4,6975	0,36721829	0,2984	1,2799379	0	1,3101496	0,0142	1,394702	0,0615	0,4997935	0,0812
0,676	1,4532	0,093855	57,4177	0,36747225	1,1399	1,0063357	0	1,0274826	0,0743	1,0866655	0,1136	0,460269	0,1826
0,617	1,2854	0,1320375	13,4063	0,35997748	1,6646	0,8921899	0	0,9098065	0,0895	0,9591095	0,1033	0,4372829	0,3198
0,571	0,9852	0,155654	14,6249	0,35087313	2,0907	0,8066863	0	0,8217741	0,0793	0,8639996	0,1501	0,4170813	0,4582
0,504	2,4351	0,1804072	28,2503	0,33250087	2,843	0,6876214	0	0,6993761	0,123	0,7322737	0,3787	0,3840832	0,6931
0,456	3,8826	0,1911088	85,9304	0,31561182	3,6364	0,6063114	0	0,6159338	0,2696	0,6428635	0,6155	0,3578368	0,8776
0,404	4,5865	0,196078	45,1689	0,29380457	5,3607	0,5219845	0	0,5295374	0,4855	0,5506755	0,9125	0,3269484	1,1142

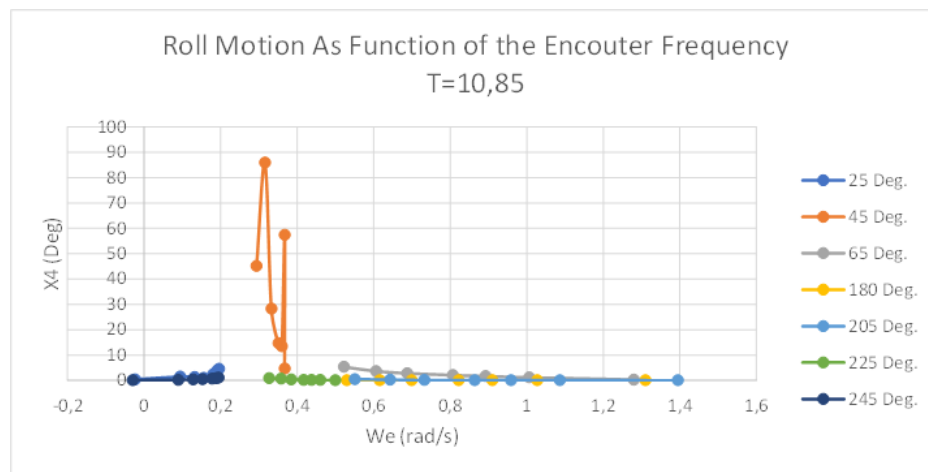


Figure 22-Roll motion as function of the encounter frequency

Derived responses and transfer functions

Included in the RAOS output file, the seakeeping tool provides both the transfer functions for each motion component and the predicted responses at selected locations—specifically, at the forward perpendicular and the bridge, which were previously defined as reference points. This section focuses on analysing how changes in vessel speed influence the behaviour of the transfer functions.

At the ship's service speed

At the vessel's service speed of 24.51 knots, the outcomes for vertical acceleration at the bridge and relative motion at the bow are illustrated in figure below.

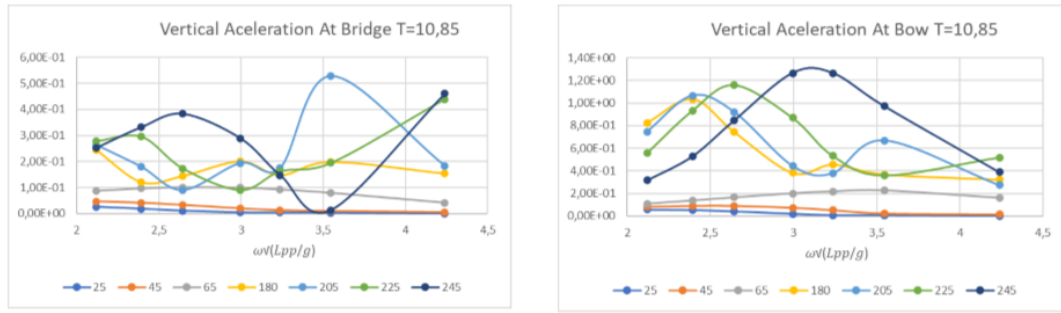


Figure 23-Vertical Acceleration at Bridge and Bow

At different ship's speeds

To assess how varying the ship's speed affects its behaviour, four different speeds—ranging from zero up to the service speed—were selected. These speeds were expressed using the nondimensional Froude number: 0.2, 0.04, 0.06, and 0.08. The resulting variations in vertical acceleration at the bridge are shown in the following graphs:

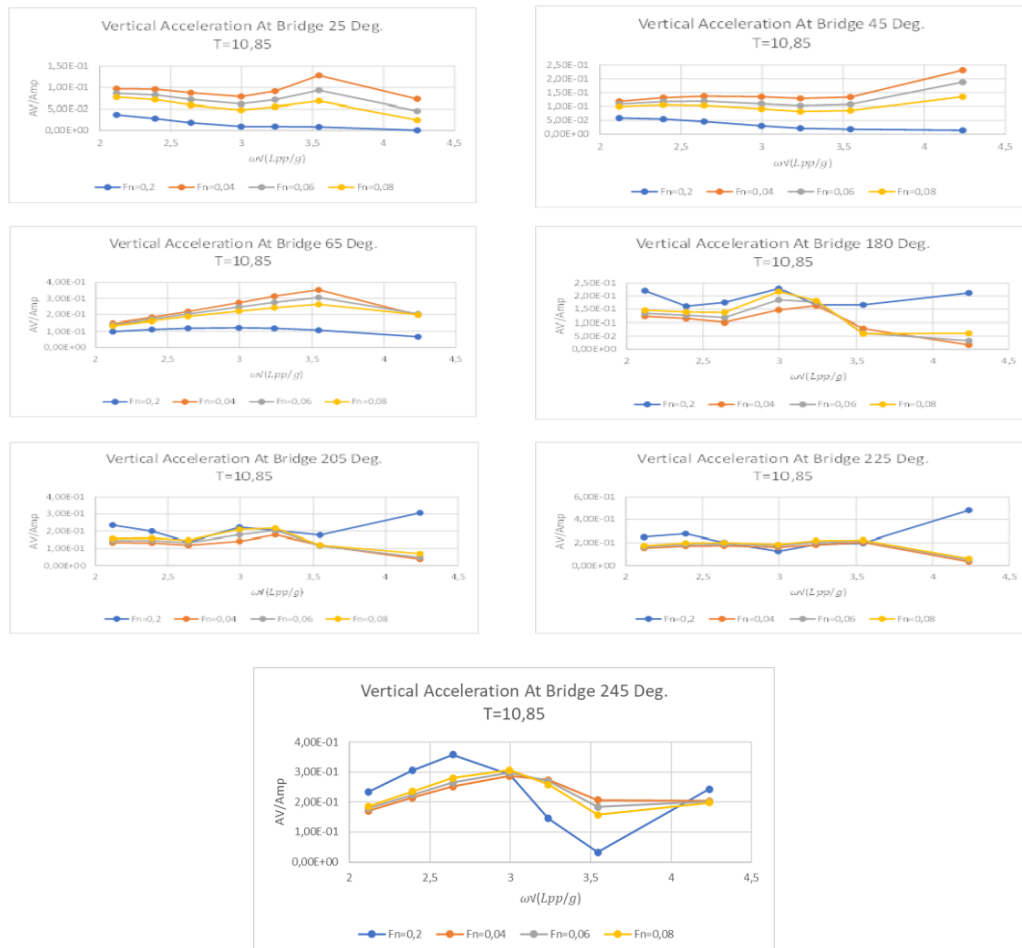


Figure 24-Vertical acceleration at the bridge for different angles

From the vertical acceleration plots under quartering seas (25°, 45°, and 65°), it can be seen that lower ship speeds consistently lead to higher accelerations across the entire frequency range. In contrast, for headings such as head, bow, and beam seas (180°, 205°, 225°, and

245°), the influence of speed is more complex: at lower frequencies, the higher speeds result in greater accelerations, while at higher frequencies, the lower speeds become more dominant.

Despite these variations, all speed conditions follow a similar trend—acceleration increases up to a peak at roughly the same frequency, followed by a decrease toward a local minimum, which also aligns across speeds. The same approach was applied to the analysis of relative motion at the bow, with the corresponding results shown below.



Figure 25-Relative motion at Bow for different angles

In the case of relative motion, all wave headings, except for 245°, exhibit a clear trend where lower ship speeds lead to greater relative displacements at the bow. The 245° case stands out as an exception to this overall pattern.

Ship motions – B44 calculated

The influence of viscous roll damping is confined to the roll motion, as this damping term is not applied to other degrees of freedom. Its presence primarily reduces the amplitude of the roll response, which is consistent with the increased energy dissipation. At smaller wave headings—specifically 25° and 45°—the viscous effect also causes noticeable shifts in the phase angle. For the other headings, its impact is mostly limited to phase variation, and nearly negligible at 205°. Even though the magnitude of the response is affected, the overall shape and trend of the response curves remain unchanged across the different scenarios.

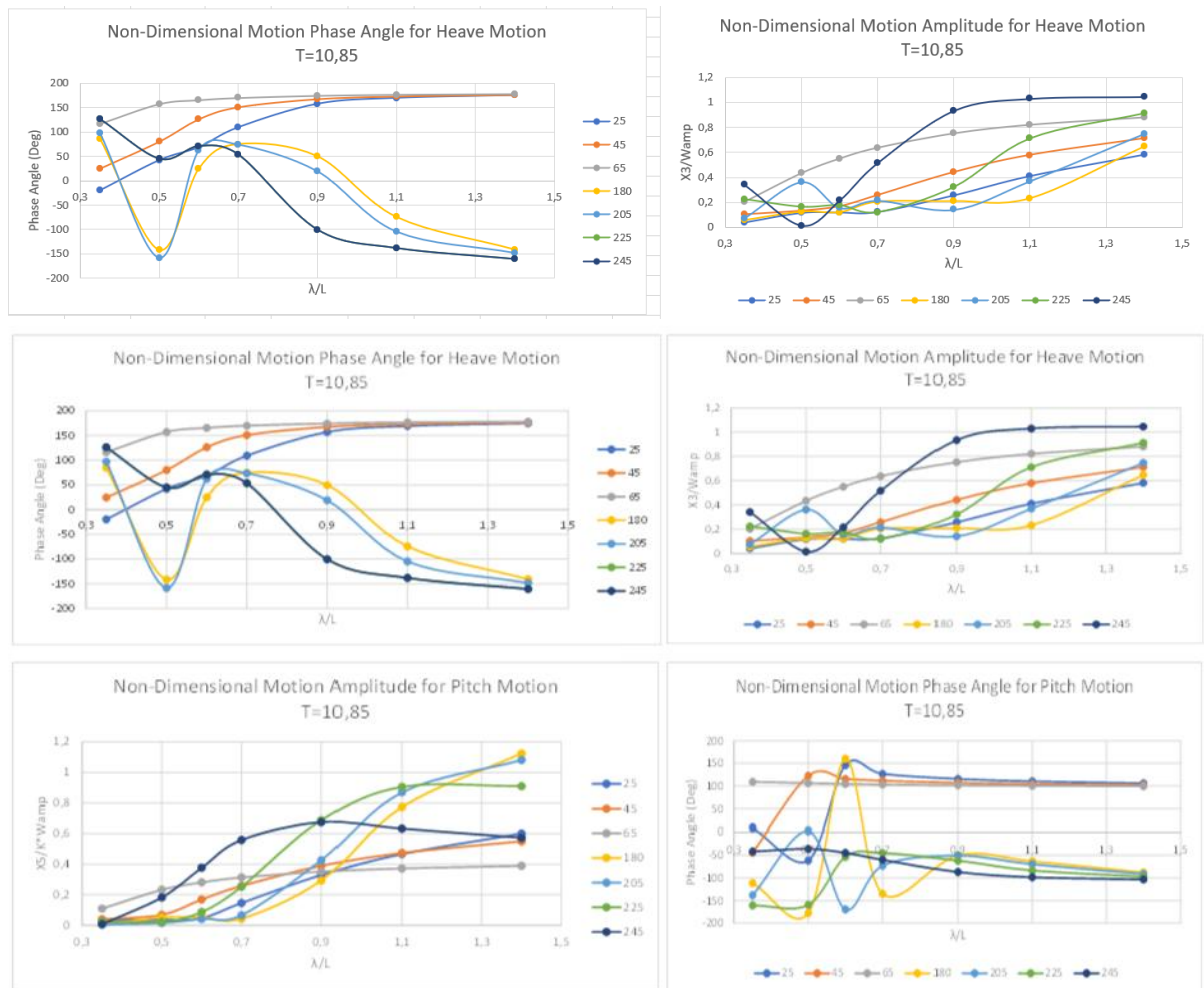


Figure 26-Effect of the viscous damping

The figures above show the graphs of heave, roll, and pitch motions generated by the seakeeping program. The roll motion amplitude graph at 45°, including the computed ψ_{44} value, is also affected by viscosity.

Roll amplitude as function of encounter frequency

Compared to the B_{44} calculated by the program, viscosity has less impact, which results in lower roll amplitude values.

Figure 27-Roll Amplitudes Table

Roll Amplitudes (Deg.)														
Wo (rad/s)	25 Deg.	We (rad/s)	45 Deg.	We (rad/s)	65 Deg.	We (rad/s)	180 Deg.	We (rad/s)	205 Deg.	We (rad/s)	225 Deg.	We (rad/s)	245 Deg.	We (rad/s)
0,808	0,4026	-0,0236881	4,3584	0,3672183	0,2965	1,27993793	0	1,3101496	0,0142	1,39470201	0,0615	0,4997935	0,0812	-0,030249
0,676	1,4413	0,09385499	6,2431	0,3674722	1,1295	1,00633573	0	1,0274826	0,0743	1,08666553	0,1136	0,460269	0,1826	0,089263
0,617	1,262	0,13203755	4,4915	0,3599775	1,6449	0,8921899	0	0,9098065	0,0895	0,95910954	0,1033	0,4372829	0,3199	0,128212
0,571	0,9554	0,15565403	6,9266	0,3508731	2,0592	0,80668632	0	0,8217741	0,0793	0,86399963	0,1501	0,4170813	0,4582	0,152377
0,504	2,3019	0,18040719	11,5276	0,3325009	2,7737	0,68762137	0	0,6993761	0,123	0,73227373	0,3787	0,3840832	0,6927	0,177854
0,456	3,6136	0,19110883	14,5134	0,3156118	3,4937	0,60631137	0	0,6159338	0,2695	0,64286353	0,6151	0,3578368	0,8763	0,189019
0,404	4,2688	0,19607796	14,8116	0,2938046	4,9098	0,52198448	0	0,5295374	0,485	0,5506755	0,9108	0,3269484	1,1099	0,194438

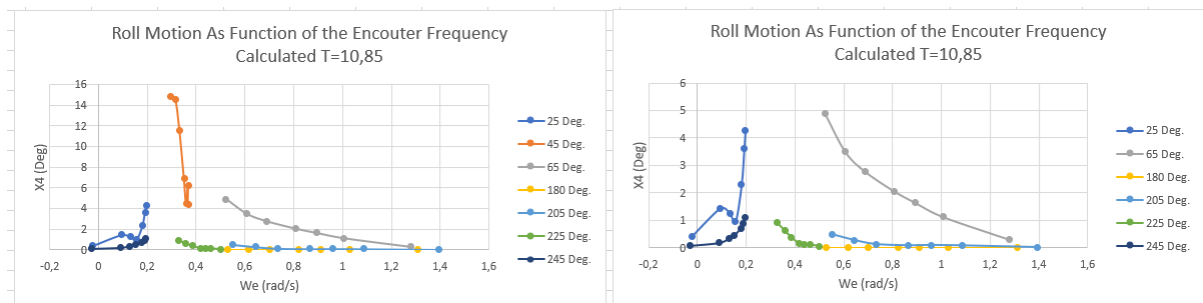


Figure 28-Roll motion amplitude as a function of the encounter frequency

Derived response and transfer functions

The following graphs will now be shown using the B_{44} values calculated.

At the ship's service speed

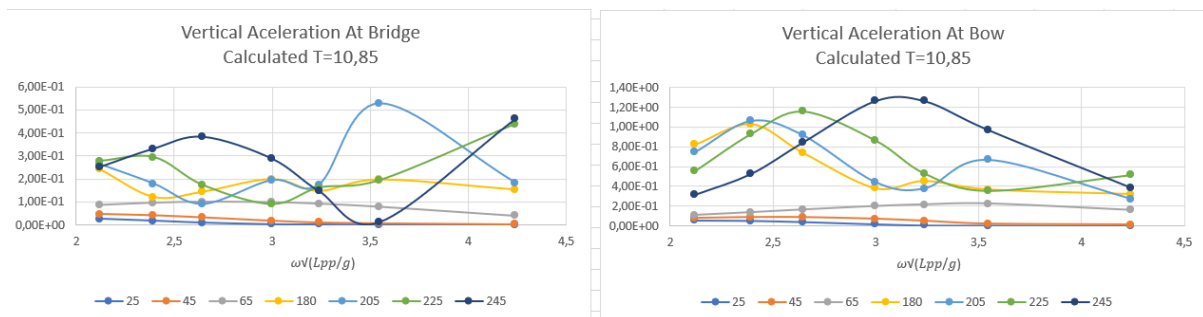


Figure 29-Vertical Acelaration at Bridge and Bow

At different ship's speeds

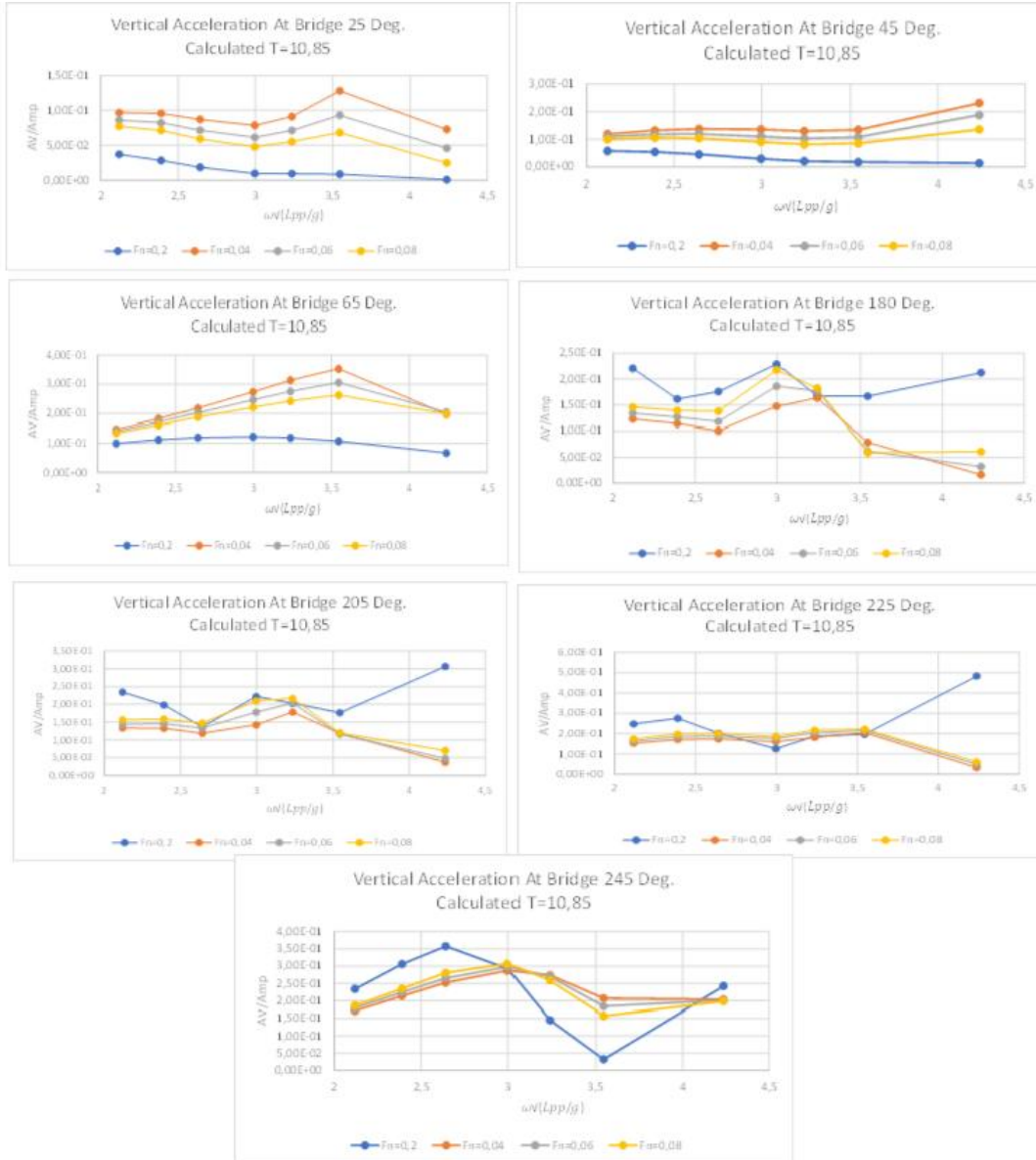


Figure 30-Vertical acceleration at the bridge



Figure 31-- Relative motion at the bow

A comparison between the vertical acceleration at the bridge and the relative motion at the bow, using both the program's *B44* and the group's calculated value, shows that the inclusion of viscosity does not lead to any visible changes in the resulting graphs.

Effect of the viscous damping coefficient (*B44 V*)

All previous calculations were based on the total roll damping coefficient, which combines both inviscid and viscous components. However, the **Solve** module within the seakeeping program also allows computations using a default *B44* value that excludes viscous effects. To evaluate the influence of the viscous damping component *B44v* on the roll motion's amplitude and phase angle, the code was rerun using this default value. For simplicity and to

summarise the comparison, the resulting graphs were produced only for a wave heading of $\beta=25^\circ$. The outcomes are presented in the following figures:

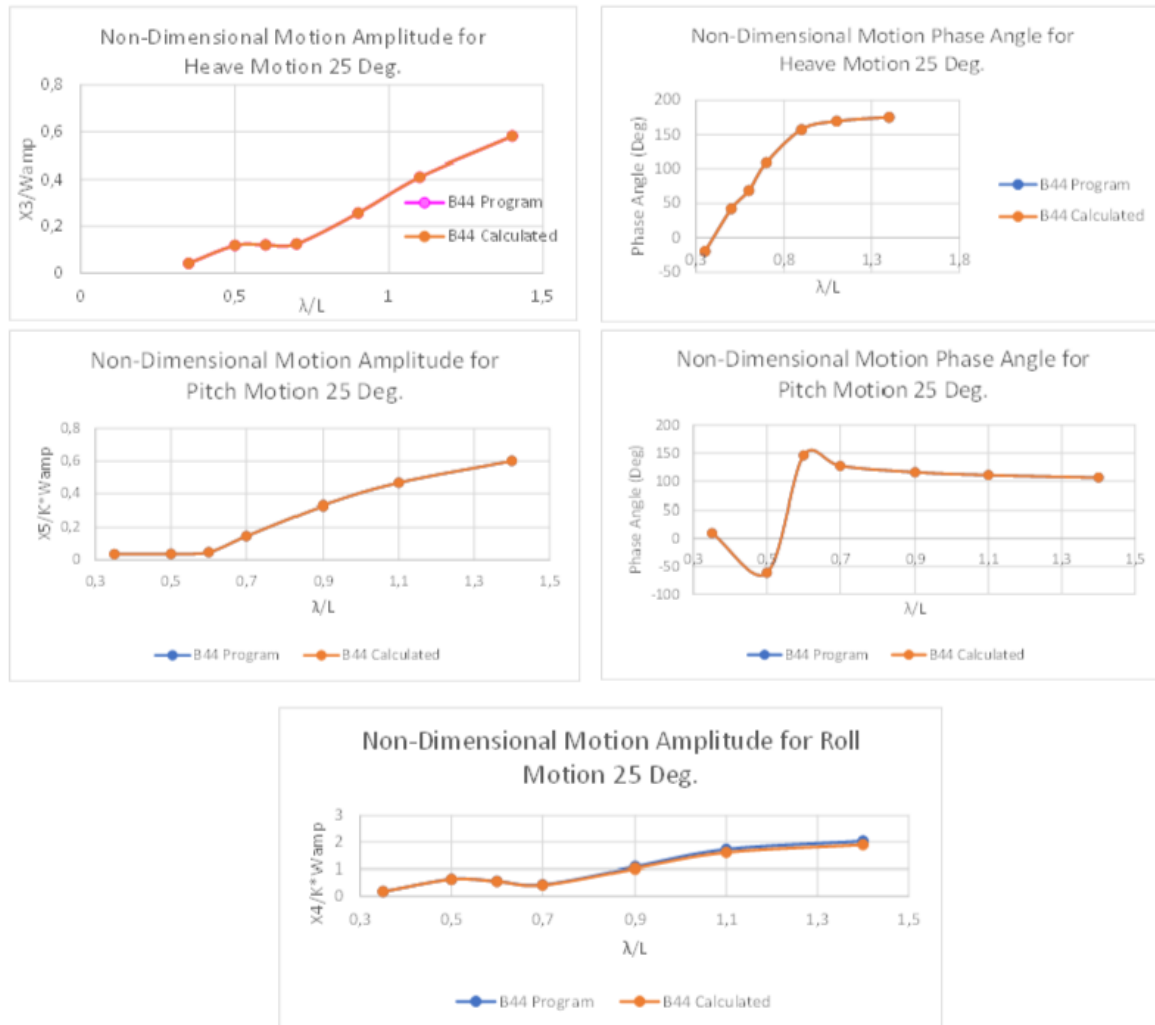
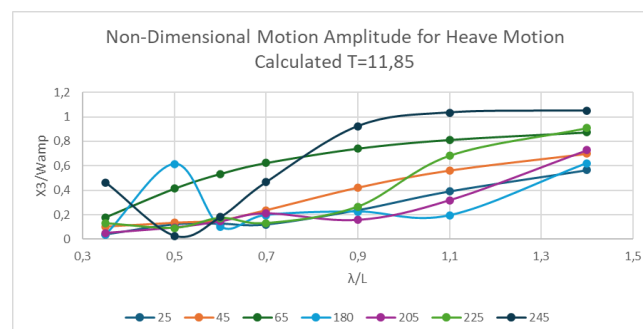


Figure 32-Non-Dimensional Motion Amplitude for Roll Motion

Difference between different waterlines

Once the calculations for the remaining two drafts were carried out, variations in Motion Amplitude for Heave Motion were observed in the MOTIOS file. As a result, the upcoming figures illustrate these differences.



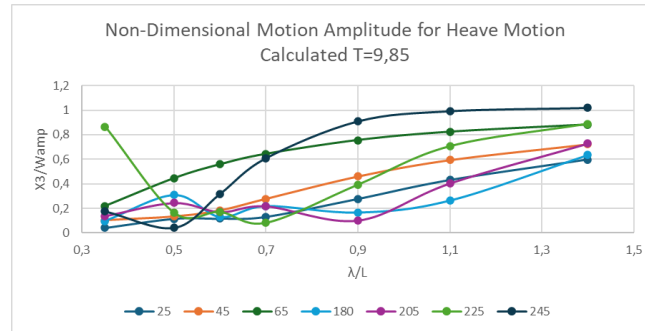


Figure 33-Nondimensional Roll motion for all heading angles

Ship Responses to Irregular Waves

Following the regular wave analysis, attention shifted to evaluating the ship's behaviour under irregular sea conditions. In this context, the sea state is irregular, stationary, and long-crested. To characterise the conditions, a scatter diagram sourced from IACS was used as a basis.

Two representative scenarios were selected. The first one reflects a more severe condition, where the significant wave height (H_s) was set at 70% of the vessel's draft, paired with the most probable peak period (T_p) for that height. The second scenario corresponds to the most frequently encountered sea state overall. Accordingly, the first case adopted $H_s = 7.5$ m and $T_p = 11.5$ s, while the second used $H_s = 1.5$ m and $T_p = 6.5$ s.

To evaluate the vessel's response under irregular wave conditions, it was necessary to perform additional computations beyond the outputs provided by the seakeeping software. The process began with the definition of the wave spectrum corresponding to the selected sea state. Based on this, the ship response spectrum was then calculated, allowing for a more

Figure 34-Figure 35-Scatter diagram from IACS.

Probability of sea-states in the North Atlantic described as occurrence per 100,000 observations																		
		Mean wave period, T_{0m1} (s)																sum
		4,5	5,5	6,5	7,5	8,5	9,5	10,5	11,5	12,5	13,5	14,5	15,5	16,5	17,5	18,5	19,5	
Significant wave height H_s (m)	0,5	6,82	202,00	333,61	187,76	45,59	4,74	0,21	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	780,73
	1,5	0,33	2028,35	12750,82	11693,39	7215,76	3006,80	846,07	160,77	20,63	1,79	0,10	0,00	0,00	0,00	0,00	0,00	37724,81
	2,5	0,00	3,38	2805,81	8517,74	7835,85	5885,37	3608,30	1805,81	737,71	246,00	66,96	14,88	2,70	0,40	0,05	0,00	31530,96
	3,5	0,00	0,00	23,06	2742,51	4666,81	4100,83	2936,41	1713,38	814,68	315,65	99,66	25,64	5,38	0,92	0,13	0,01	17445,07
	4,5	0,00	0,00	0,00	82,06	1759,81	2069,19	1715,42	1151,29	625,51	275,12	97,96	28,24	6,59	1,24	0,19	0,02	7812,64
	5,5	0,00	0,00	0,00	0,08	149,74	811,81	791,81	609,66	375,67	185,26	73,12	23,09	5,84	1,18	0,19	0,02	3027,47
	6,5	0,00	0,00	0,00	0,00	1,02	147,59	305,37	271,71	190,23	104,79	45,42	15,49	4,16	0,88	0,15	0,02	1086,83
	7,5	0,00	0,00	0,00	0,00	0,00	4,77	88,62	107,20	86,26	53,35	25,36	9,27	2,60	0,56	0,09	0,01	378,09
	8,5	0,00	0,00	0,00	0,00	0,00	0,02	9,40	38,70	36,80	25,95	13,63	5,33	1,55	0,34	0,05	0,01	131,78
	9,5	0,00	0,00	0,00	0,00	0,00	0,00	0,20	9,34	15,15	12,51	7,39	3,12	0,94	0,20	0,03	0,00	48,88
	10,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,81	5,73	5,96	4,08	1,90	0,60	0,13	0,02	0,00	19,23
	11,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,02	1,29	2,68	2,23	1,18	0,40	0,08	0,01	0,00	7,89
	12,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,11	1,01	1,14	0,72	0,27	0,06	0,01	0,00	3,32
	13,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,22	0,51	0,42	0,18	0,04	0,00	0,00	1,37
	14,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,02	0,19	0,21	0,12	0,03	0,00	0,00	0,57
	15,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,04	0,09	0,07	0,02	0,00	0,00	0,22
	16,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,03	0,04	0,01	0,00	0,00	0,08
	17,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,01	0,02	0,01	0,00	0,00	0,04
	18,5	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,01	0,01	0,01	0,00	0,02
SUM		7,15	2233,73	15913,30	23223,54	21674,58	16031,12	10301,81	5868,69	2909,77	1230,31	437,79	129,62	31,47	6,11	0,92	0,09	100000,00

Variance spectrum of the sea state

In wave analysis, the concept of variance is used to describe how the elevation of the sea surface changes over time. The variance spectrum distributes this variability across different frequencies, essentially showing how wave energy is spread throughout the frequency range of a given sea state.

By interpreting this spectrum, it's possible to gather meaningful information about the sea conditions at a specific location, which can then be applied to predict how a vessel might behave when exposed to those waves. For this purpose, variance spectra were produced for both sea states under consideration. These were expressed with respect to both the wave frequency (ω) and the encounter frequency (ω_e), which accounts for the ship's motion relative to the waves.

To define the energy distribution in terms of wave frequency, the ISSC spectrum was chosen. This model is appropriate for fully developed seas and describes how the energy density varies with frequency. Its mathematical formulation is given by:

$$S(\omega) = \frac{0.313 \cdot H_s^2 \cdot \omega_p^4}{\omega^5} \cdot \exp\left(-1.25 \cdot \left(\frac{\omega_p}{\omega}\right)^4\right) \quad [m^2 \cdot s]$$

Where,

- H_s is the significant wave height;
- ω_p is the peak wave period given by 2π .

To express the wave spectrum in terms of the encounter frequency (ω_e), a transformation was applied to account for the relative motion between the ship and the incoming waves. This is particularly important when the vessel is advancing with constant speed and encountering waves from a certain angle (β). The transformation is given by:

$$\omega_e = \omega - \frac{\omega^2 V_s \cos(\beta)}{g}$$
$$S(\omega_e, \beta) = \frac{S(\omega)}{1 - \frac{2\omega U \cos \beta}{g}} \quad [m^2 \cdot s]$$

Where,

- V_s is the ship velocity in [m/s];
- β is the heading angle in [deg];
- g is the gravity acceleration in [m/s²].

$H_s = 70\%$ of the initial draft

In this analysis, the ship's response is evaluated under head wave conditions. The selected sea state corresponds to 70% of the vessel's initial draft, which is 10.85 m.

This results in a significant wave height $H_s = 7.5$ m and a peak period $T_p = 11.5$ s. Based on these parameters, the following figures illustrate the computed results for the ship's behaviour in this wave condition.

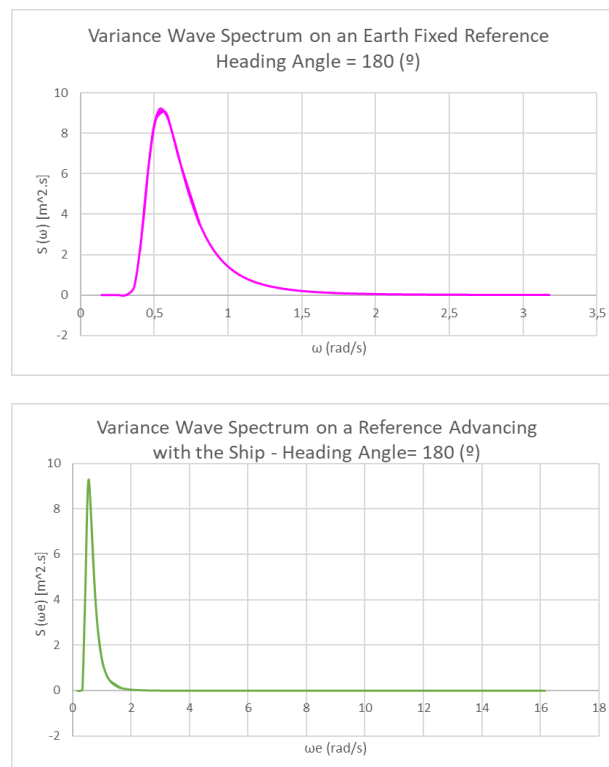
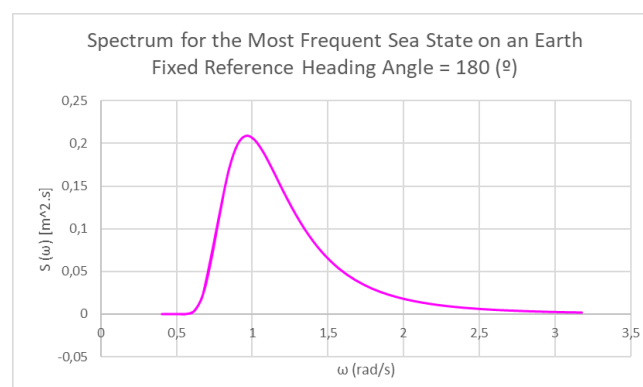


Figure 35-Variance Wave Spectrum

Most frequent sea state

For the most common sea state, defined by a significant wave height of $H_s = 1.5$ m and a peak period $T_p = 6.5$ s, the ship's response was evaluated under head wave conditions. The results obtained for this scenario are presented in the following figures:



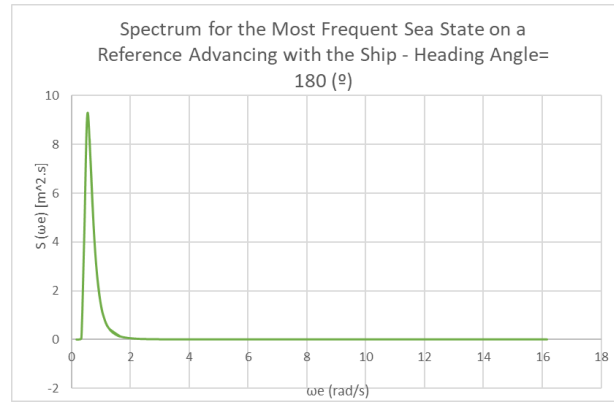


Figure 36-Spectrum for the Most frequent Sea State on a Reference Advancing with the ship

Response spectra

With the variance spectra now defined, it becomes possible to compute the response spectra corresponding to the sea state under analysis. This is achieved by incorporating the previously obtained transfer function into the wave spectrum, allowing for the evaluation of how each wave component influences the ship's motion. The response spectrum is calculated using the following expression:

$$S_j(\omega, \beta) = |x_a^j(\omega, \beta)|^2 S_\xi(\omega)$$

Where:

- $S_j(\omega, \beta)$ is the spectrum response;
- $|x_a^j(\omega, \beta)|^2$ is the amplitude of the transfer function;
- $S_\xi(\omega)$ is the wave spectrum.

Heave

The following graph presents the response spectrum obtained for the heave motion:

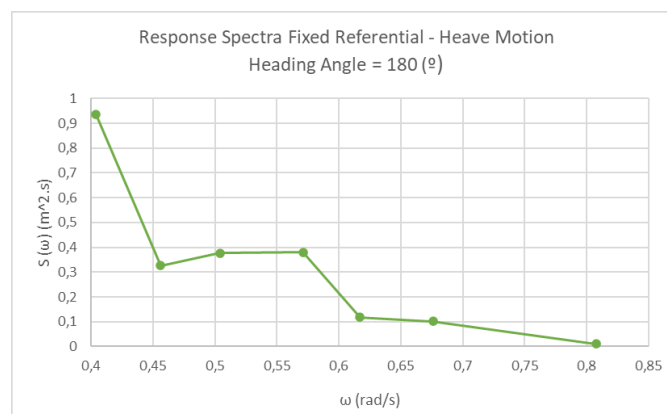


Figure 37-Response Spectra Fixed Referential

Pitch

The graph below shows the response spectrum for pitch motion:

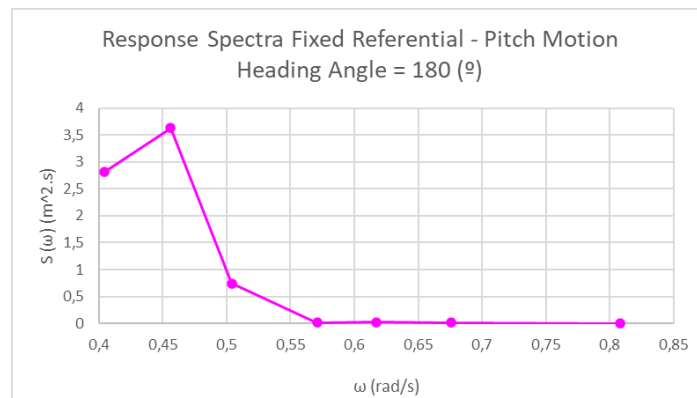


Figure 38-Response Spectra Fixed Referential

Vertical Acceleration at the Bridge

The following graph illustrates the response spectrum of the vertical acceleration at the bridge, providing insight into the dynamic loads experienced in this region under wave excitation.

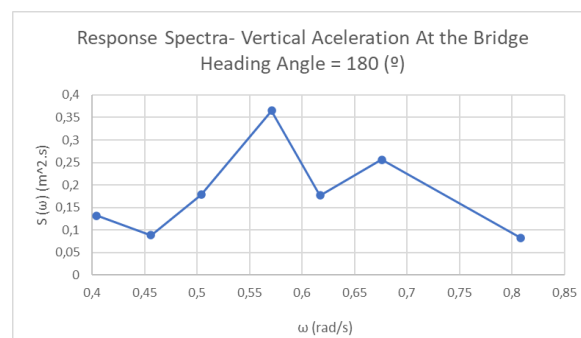


Figure 39-Response Spectra-Vertical acceleration at the bridge

Relative Movement of the Bow

The graph below displays the response spectrum for the relative movement at the bow, capturing the motion between the ship's structure and the wave surface.

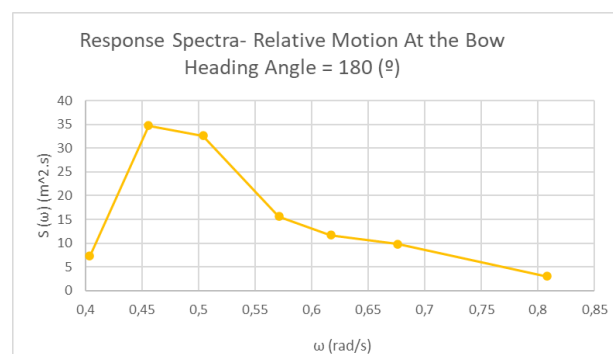


Figure 40-Response Spectra-Relative motion at the bow

Rayleigh distributions

One way to study the short-term behavior of the previous responses is through examining the distribution of their peak values. These maxima are often well-represented by a Rayleigh distribution, which can be expressed by the equation below:

$$p(\xi) = \frac{\xi}{E} \exp\left(-\frac{\xi^2}{2E}\right)$$

In the equation above, E corresponds to the sea state variance, which is equivalent to the total area beneath the variance spectrum. The variable ξ refers to a particular wave height. The spectrum used in this context is the ship's response spectrum.

The results obtained from applying the Rayleigh distribution are shown below:

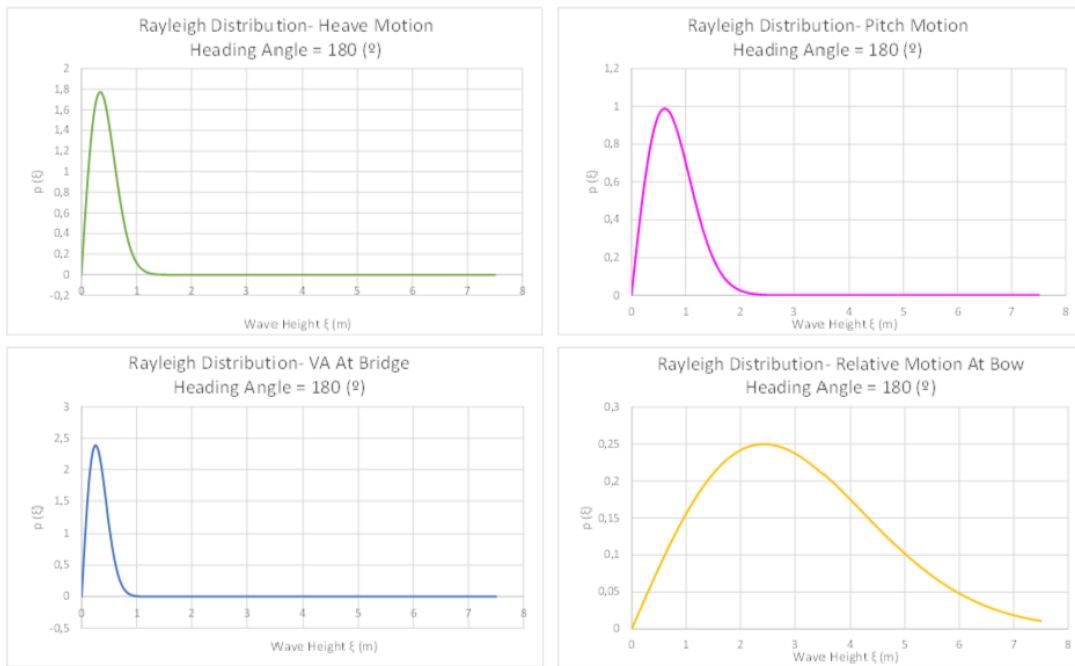


Figure 41-Rayleigh Distribution

Probability of Heave Motion Higher than 3 m

To determine the probability of a heave motion exceeding 3 meters, the following expression was used:

$$p(x_j > x_j^*) = \exp\left(-\frac{(x_j^*)^2}{2E}\right)$$

In this case, $x_j^* = 3$ m, and E corresponds to the variance of the sea state. The resulting probability was zero, indicating that such extreme vertical motion is highly unlikely under the given conditions.

Conclusion

The work carried out throughout this project has enabled a comprehensive exploration of ship dynamics in wave environments, grounded in the concepts addressed during the *Dynamics and Hydrodynamics of Floating Bodies* course. Focusing on ship behaviour under various sea states, the study made use of strip theory to simulate and interpret vessel motions, offering valuable conclusions about their seakeeping performance.

A seakeeping analysis tool was employed to estimate these motions, and despite its overall usefulness, some divergences were noted when cross-checked with the results from the reference study. These differences suggest that refinements—such as modifying the number of longitudinal sections—could potentially increase result fidelity. It should be highlighted that some of these mismatches may stem from simplifications inherent in the software's theoretical framework. A specific issue was identified in roll motion at a 45° heading, where unexpected errors prevented any solid conclusions for that condition.

In the context of irregular wave modelling, a significant wave height of 8.5 m was adopted. Although the value derived from the scatter diagram, based on 70% of the ship's draft, was closer to 7.5 m, the higher estimate was selected in line with to ensure a more conservative analysis.

While simulations were carried out for different drafts, only the results for a waterline level of 10.85 m are shown in this document. Differences observed in roll motion phase angles across various drafts justify the exclusion of plots for those alternative conditions.

Altogether, this study not only reinforced the theoretical foundation laid out in class but also expanded the analytical understanding of how ships respond to wave-induced forces. It paves the way for more advanced work in seakeeping, highlighting the importance of ongoing development and validation of computational tools in naval architecture.

The knowledge and skills acquired through this research are expected to be of significant value in future applications, whether in ship design, performance assessment, or the broader field of maritime safety and engineering.

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